

- 4 A hydrogen atom consists of a proton and an electron separated by distance $2x$, as shown in Figure 4.1.

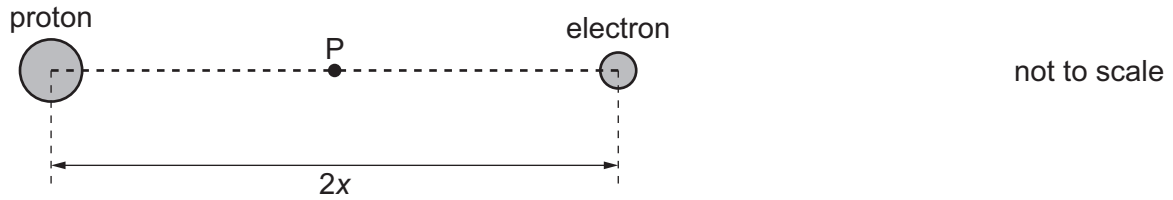


Figure 4.1

Point P is at the midpoint of the line joining the centres of the proton and the electron.

- (a) (i) For point P, compare the electric field due to the proton with the electric field due to the electron.

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..... [3]

- (ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron.

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- (b) (i) The elementary charge is e and the permittivity of free space is ϵ_0 .

State an expression, in terms of x , e and ϵ_0 , for the magnitude of the electric force F between the proton and the electron.

$F =$ [2]

- (ii) Determine an expression, in terms of F and x , for the electric potential energy E_p of the proton and the electron.

$$E_p = \dots\dots\dots [1]$$

[Total: 9]